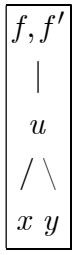
Solutions Tutorial 3

1. Let $f:\mathbb{R}\to\mathbb{R}$ be a twice differentiable. Find the expressions for h_x, h_y and h_{xy} in terms of f' and f'' for the following 2-variable functions:

(a) $h(x,y)=xf(x^2-y)$ (b) $h(x,y)=\big(f(xy)\big)^2$ (c) $h(x,y)=\frac{x+f(y)}{f(x)}$ (d) $h(x,y)=xe^{yf(x)}$

Solution. (a) $h(x,y)=xf(x^2-y)=xf(u),\ u=x^2-y,\ u_x=2x,\ u_y=-1$

$$\Rightarrow\begin{cases}h_x=f(u)+x\frac{\partial}{\partial x}f(u)=f(u)+xf'(u)\times u_x=f(u)+2x^2f'(u)\\ h_y=x\frac{\partial}{\partial y}f(u)=xf'(u)\times u_y=xf'(u)\times(-1)=-xf'(u)\\ h_{xy}=-\frac{\partial}{\partial x}\big(xf'(u)\big)=-f'(u)-x\frac{\partial}{\partial x}f'(u)=-f'(u)-2x^2f''(u)\end{cases}$$



(b) $h(x,y)=\big(f(xy)\big)^2=\big(f(u)\big)^2,\ u=xy$ (Note that h is a symmetric function)

$$\Rightarrow\begin{cases}h_x=2f(u)f'(u)u_x=2f(u)f'(u)y\\ h_y=2f(u)f'(u)u_y=2f(u)f'(u)x\end{cases}$$

Take natural log of h_x on both sides and partially differentiate it with respect to y:

$$\ln h_x=\ln(2y)+\ln f(u)+\ln f'(u)$$
$$\Rightarrow\underset{\frac{\partial}{\partial y}}{\frac{h_{xy}}{h_x}}=\frac{2}{2y}\frac{\partial y}{\partial y}+\frac{1}{f(u)}\frac{\partial f(u)}{\partial y}+\frac{1}{f'(u)}\frac{\partial f'(u)}{\partial y}=\frac{1}{y}+\frac{1}{f(u)}f'(u)\times u_y+\frac{1}{f'(u)}f''(u)\times u_y$$
$$\Rightarrow h_{xy}=2yf(u)f'(u)\Big(\frac{1}{y}+\frac{xf'(u)}{f(u)}+\frac{xf''(u)}{f'(u)}\Big)$$

(c) $h(x,y)=\frac{x+f(y)}{f(x)}\Rightarrow\begin{cases}h_x=\frac{1\times f(x)-f'(x)(x+f(y))}{(f(x))^2}=\frac{f(x)-f'(x)(x+f(y))}{(f(x))^2}\\ h_y=\frac{f'(y)f(x)-0}{(f(x))^2}=\frac{f'(y)}{f(x)}\\ h_{xy}=\frac{0-f'(x)f'(y)}{(f(x))^2}\end{cases}$

(d) $h(x,y)=xe^{yf(x)}\Rightarrow\begin{cases}h_x=e^{yf(x)}+xyf'(x)e^{yf(x)}\\ h_y=xf(x)e^{yf(x)}\end{cases}$

Take natural log of h_y on both sides and partially differentiate it with respect to x:

$$\ln h_y=\ln x+\ln f(x)+yf(x)\Rightarrow h_{xy}=h_{yx}=xf(x)e^{yf(x)}\Big(\frac{1}{x}+\frac{f'(x)}{f(x)}+yf'(x)\Big).$$

Remark. Using the chain rule $\frac{d}{dx}\ln(f(x))=\frac{f'(x)}{f(x)}$, we can find a nice formula to find the derivative involving product of functions. Let $f(x)=u(x)v(x)w(x)$. Then

Thus,
$$\ln f(x)=\ln u(x)+\ln v(x)+\ln w(x).$$
$$\frac{d}{dx}\ln f(x)=\frac{f'(x)}{f(x)}=\frac{u'(x)}{u(x)}+\frac{v'(x)}{v(x)}+\frac{w'(x)}{w(x)}$$
$$\Rightarrow f'(x)=f(x)\Big(\frac{u'(x)}{u(x)}+\frac{v'(x)}{v(x)}+\frac{w'(x)}{w(x)}\Big)$$

In general,
$$f(x)=\prod_{i=1}^{n}f_i(x)\Rightarrow f'(x)=f(x)\sum_{i=1}^{n}\frac{f_i'(x)}{f_i(x)}.$$

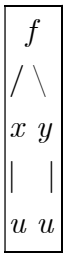
2. Let $f(x,y)$ be a C^1-class function of two variables.

(a) Express h' in terms of partial derivatives of the function f:

(i) $h(u)=f(u,e^u)$ (ii) $h(u)=\Big(f(u,u)\Big)^2$ (iii) $h(u)=\sin\Big(f(2u,3u^2)\Big)$

(b) For (a)(i), express h'' in terms of partial derivatives of f.

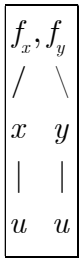
Solution. (a)(i) $h'(u)=f_x(u,e^u)\times\frac{dx}{du}+f_y(u,e^u)\times\frac{dy}{du}=f_x(u,e^u)+e^uf_y(u,e^u)$



(ii) $h'(u)=2f(u,u)\Big(f_x(u,u)\times\frac{dx}{du}+f_y(u,u)\times\frac{dy}{du}\Big)$
$=2f(u,u)\Big(f_x(u,u)+f_y(u,u)\Big)$

(iii) $h'(u)=\cos\Big(f(2u,3u^2)\Big)\times f_x(u,u)\times\frac{dx}{du}+\frac{dh}{df}\times f_y(u,u)\times\frac{dy}{du}$
$=\cos\Big(f(2u,3u^2)\Big)\Big(2f_x(2u,3u^2)+6uf_y(2u,3u^2)\Big)$

(b) Let $x=u,\ y=e^u$. Then $\frac{dx}{du}=1,\ \frac{dy}{du}=e^u$. So,



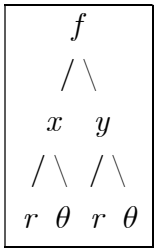
$h''(u)=\frac{d}{du}\Big(f_x(u,e^u)\Big)+\frac{d}{du}\Big(e^uf_y(u,e^u)\Big)$
$=\frac{d}{du}\Big(f_x(u,e^u)\Big)+e^u\frac{d}{du}\Big(f_y(u,e^u)\Big)+e^uf_y(u,e^u)$
$=f_{xx}\times\frac{dx}{du}+f_{xy}\times\frac{dy}{du}+e^u\left(f_{yx}\times\frac{dx}{du}+f_{yy}\times\frac{dy}{du}\right)+e^uf_y$
$=f_{xx}+e^uf_{xy}+e^u\left(f_{yx}+e^uf_{yy}\right)+e^uf_y$
$=f_{xx}+2e^uf_{xy}+e^{2u}f_{yy}+e^uf_y$

3. Let $(x,y)=(r\cos\theta,r\sin\theta)$, $f(x,y)=x^2+xy-y^2$ and $h(r,\theta)=f(x(r,\theta),y(r,\theta))$.
Find $\frac{\partial h}{\partial r}$ and $\frac{\partial h}{\partial\theta}$ by (i) direct substitution; (ii) the chain rule.

Solution. (i) $h(r,\theta)=r^2\Big(\cos^2\theta+\cos\theta\sin\theta-\sin^2\theta\Big)=r^2\Big(\cos2\theta+\frac12\sin2\theta\Big)$.

Then $\frac{\partial h}{\partial r}=2r\Big(\cos2\theta+\frac12\sin2\theta\Big)$ and $\frac{\partial h}{\partial\theta}=r^2\Big(-2\sin2\theta+\cos2\theta\Big)$.

(ii)



$\frac{\partial h}{\partial r}=\frac{\partial f}{\partial x}\times\frac{\partial x}{\partial r}+\frac{\partial f}{\partial y}\times\frac{\partial y}{\partial r}=(2x+y)\cos\theta+(x-2y)\sin\theta$
$=r(2\cos\theta+\sin\theta)\cos\theta+r(\cos\theta-2\sin\theta)\sin\theta$
$=2r(\cos^2\theta+\sin\theta\cos\theta-\sin^2\theta)=2r\Big(\cos2\theta+\frac12\sin2\theta\Big)$

$\frac{\partial h}{\partial\theta}=\frac{\partial f}{\partial x}\times\frac{\partial x}{\partial\theta}+\frac{\partial f}{\partial y}\times\frac{\partial y}{\partial\theta}=-r(2x+y)\sin\theta+r(x-2y)\cos\theta$
$=-r^2(2\cos\theta+\sin\theta)\sin\theta+r^2(\cos\theta-2\sin\theta)\cos\theta$
$=r^2(\cos^2\theta-4\sin\theta\cos\theta-\sin^2\theta)=r^2\Big(\cos2\theta-2\sin2\theta\Big)$

4. Let $G(u,v) = f\Big(x(u,v), y(u,v)\Big)$, where $f(x,y), x(u,v)$ and $y(u,v)$ are functions of two variables. Compute $G_u(0,0)$ if

$$x(0,0) = 2\,,\ y(0,0) = -1\,,\ x_u(0,0) = 7\,,\ y_u(0,0) = -2\,,\ f_x(2,-1) = 0\,,\ f_y(2,-1) = \tfrac{1}{2}\,.$$

Solution. $\frac{\partial}{\partial u} f(x(u,v), y(u,v)) = f_x(x(u,v), y(u,v)) \times x_u(u,v) + f_y(x(u,v), y(u,v)) \times y_u(u,v)\,.$

$$\begin{array}{c} f \\ / \ \backslash \\ x \quad y \\ / \backslash \quad / \backslash \\ u \quad v \quad u \quad v \end{array}$$

$$\begin{aligned} G_u(0,0) &= \tfrac{\partial}{\partial u} f(x(0,0), y(0,0)) \\ &= f_x(x(0,0), y(0,0)) \times x_u(0,0) + f_y(x(0,0), y(0,0)) \times y_u(0,0) \\ &= f_x(2,-1) \times x_u(0,0) + f_y(2,-1) \times y_u(0,0) \\ &= 0 \times 2 + \tfrac{1}{2} \times (-2) \\ &= -1 \end{aligned}$$

5. Let $f(u,v) : \mathbb{R}^2 \to \mathbb{R}$ be a function with $(u,v) = (2x - y, 3x + 2y)$. Find formulae for $\frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial y \partial x}$ in terms of partial derivatives of f.

Solution. $u_x = 2,\ u_y = -1,\ v_x = 3,\ v_y = 2\,.$

$$\begin{array}{c} f, f_u, f_v \\ / \ \backslash \\ u \quad v \\ / \backslash \quad / \backslash \\ x \quad y \quad x \quad y \end{array}$$

$$\begin{aligned} \tfrac{\partial f}{\partial y} &= f_u u_y + f_v v_y = -f_u + 2f_v \\ \tfrac{\partial f}{\partial y \partial x} &= \tfrac{\partial}{\partial x}(-f_u + 2f_v) \\ &= -\Big(f_{uu} u_x + f_{uv} v_x\Big) + 2\Big(f_{vu} u_x + f_{vv} v_x\Big) \\ &= -\Big(2f_{uu} + 3f_{uv}\Big) + 2\Big(2f_{vu} + 3f_{vv}\Big) \\ &= -2f_{uu} + f_{uv} + 6f_{vv} \end{aligned}$$

6. Let $f(x,y) : \mathbb{R}^2 \to \mathbb{R}$ be a function such that $(x,y) = (r\cos\theta, r\sin\theta)\,.$

Let $h(r,\theta) = f(x(r,\theta), y(r,\theta))\,.$ Show that $h_r^2 + \frac{1}{r^2} h_\theta^2 = f_x^2 + f_y^2\,.$

Solution. $h_r = f_x x_r + f_y y_r = f_x \cos\theta + f_y \sin\theta\,, h_\theta = f_x x_\theta + f_y y_\theta = r\Big(-f_x \sin\theta + f_y \cos\theta\Big).$

Therefore, $h_r^2 + \frac{1}{r^2} h_\theta^2 = \Big(f_x \cos\theta + f_y \sin\theta\Big)^2 + \Big(-f_x \sin\theta + f_y \cos\theta\Big)^2 = f_x^2 + f_y^2\,.$

Remark: One can also write $f_r^2 + \frac{1}{r^2} f_\theta^2 = f_x^2 + f_y^2\,.$

7. Let $f(u,v), h(x,y) : \mathbb{R}^2 \to \mathbb{R}$ be two 2-variable functions, where $(u,v) = \Big(xy, \frac{x}{y}\Big)$ and $h(x,y) = f(u(x,y), v(x,y))\,.$ Show that $xh_x + yh_y = 2uf_u$ and $xh_x - yh_y = 2vf_v\,.$

Solution.

$$\begin{aligned} h_x &= f_u u_x + f_v v_x = y f_u + \tfrac{1}{y} f_v = y\Big(f_u + \tfrac{1}{y^2} f_v\Big) \Rightarrow \tfrac{1}{y} h_x = f_u + \tfrac{1}{y^2} f_v \quad (1) \\ h_y &= f_u u_y + f_v v_y = x f_u - \tfrac{x}{y^2} f_v = x\Big(f_u - \tfrac{1}{y^2} f_v\Big) \Rightarrow \tfrac{1}{x} h_y = f_u - \tfrac{1}{y^2} f_v \quad (2) \end{aligned}$$

$$\begin{aligned} (1) + (2) &: \tfrac{1}{y} h_x + \tfrac{1}{x} h_y = 2f_u \underset{\times xy}{\Rightarrow} xh_x + yh_y = 2xyf_u = 2uf_u \\ (1) - (2) &: \tfrac{1}{y} h_x - \tfrac{1}{x} h_y = 2y^{-2} f_v \underset{\times xy}{\Rightarrow} xh_x - yh_y = \tfrac{2x}{y} f_v = 2vf_v \end{aligned}.$$

8. Find the directional derivative for the functions

$$\begin{cases} f(x,y) = x^3 y + e^x & \text{at } (x,y) = (1,1) \\ g(x,y) = xe^{2x+3y} & \text{at } (x,y) = (1,0) \end{cases}$$

in the direction defined by the vector $v = (1,1)$.

Solution. $\widehat{v} = \frac{1}{\sqrt{2}}(1,1)$. Then

$$\begin{cases} \nabla f(x,y) = (3x^2 y + e^x, x^3) & \Rightarrow \nabla f(1,1) = (3+e,1) \\ \nabla g(x,y) = (e^{2x+3y}(1+2x), 3xe^{2x+3y}) & \Rightarrow \nabla g(1,0) = (3e^2, 3e^2) \end{cases}$$

So $$\begin{cases} \partial_{(1,1)} f(1,1) = \frac{1}{\sqrt{2}}(1,1)\cdot(3+e,1) = \frac{4+e}{\sqrt{2}} \\ \partial_{(1,1)} g(1,0) = \frac{1}{\sqrt{2}}(1,1)\cdot(3e^2, 3e^2) = \frac{6e^2}{\sqrt{2}} \end{cases}$$

Extra Exercises

9. Write $\frac{\partial h}{\partial x}$ if $h(x,y)$ equals to

(a) $f(x,u(x,y))$ (b) $f(x,u(x),v(x))$ (c) $f(u(x,y,z),v(x,y),w(x))$

(d) $f(x,yu(x,y))$ (e) $f(xu(x),yv(x,y))$

Solution.

(a)	f $/ \quad \backslash$ $x \quad u$ $/ \backslash$ $x \quad y$	Given $f(x,u(x,y))$. $\frac{\partial h}{\partial x} = f_x + f_u u_x$
(b)	f $/ \mid \backslash$ $x \quad u \quad v$ $\mid \quad \mid$ $x \quad x$	Given $f(x,u(x),v(x))$. $\frac{\partial h}{\partial x} = f_x + f_u \frac{du}{dx} + f_v \frac{dv}{dx}$
(c)	f $/ \quad \mid \quad \backslash$ $u \quad v \quad w$ $/\mid\backslash \quad /\backslash \quad \mid$ $x\ y\ z\ x\ y \quad x$	Given $f(u(x,y,z),v(x,y),w(x))$. $\frac{\partial h}{\partial x} = f_u u_x + f_v v_x + f_w \frac{dw}{dx}$
(d)	f $/ \quad \backslash$ $x \quad w$ $/ \backslash$ $x \quad y$	Given $f(x,yu(x,y))$. Let $w = yu(x,y) \Rightarrow w_x = yu_x$. $h_x = f_x + f_w w_x = f_x + f_w yu_x$
(e)	f $/ \quad \backslash$ $s \quad t$ $\mid \quad / \backslash$ $x \quad x \quad y$	Given $f(xu(x),yv(x,y))$. Let $s = xu(x), t = yv(x,y)$. Then $s' = u + xu', t_x = yv_x$. $h_x = f_s s' + f_t t_x = f_s \cdot (u + xu') + f_t \cdot yv_x$

10. Let $w = f(x,y,t)$ with x and y depending on t. Suppose at point (x, y) and at time t, the partial derivatives f_x, f_y and f_t are equal to 2, –3 and 5 respectively, while $\frac{dx}{dt} = 1$ and $\frac{dy}{dt} = 2$. Find and explain the difference between $\frac{dw}{dt}$ and f_t.

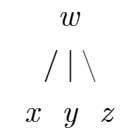
Solution. $\frac{dw}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} + f_t = 2(1) - 3(2) + 5 = 1$.

f_t gives the rate of change of w as t varies while x and y are fixed; while $\frac{dw}{dt}$ gives the rate of change of w as all $x(t)$, $y(t)$ and t vary at the same time.

11. Use two methods (one using the chain rule) to evaluate w_s and w_t given that the function $w = x^2 + y^2 + z^2$, with $x = st$, $y = s\cos t$, and $z = s\sin t$.

Solution. Method 1 substitution: $w = x^2 + y^2 + z^2 = s^2t^2 + s^2\cos^2 t + s^2\sin^2 t$

$$\Rightarrow \begin{cases} w_s = 2st^2 + 2s\cos^2 t + 2s\sin^2 t = 2s(t^2+1) \\ w_t = 2s^2 t - 2s^2\cos t\sin t + 2s^2\sin t\cos t = 2s^2 t \end{cases}$$



Method 2 chain rule: $\begin{cases} w_x = 2x \\ w_y = 2y \\ w_z = 2z \end{cases}$, $\begin{cases} x_s = t, & x_t = s. \\ y_s = \cos t, & y_t = -s\sin t, \\ z_s = \sin t, & z_t = s\cos t. \end{cases}$

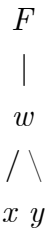
Thus, $w_s = w_x x_s + w_y y_s + w_z z_s$
$= 2xt + 2y\cos t + 2z\sin t$
$= 2st^2 + 2s\cos^2 t + 2s\sin^2 t$
$= 2s(t^2+1)$

$w_t = w_x x_t + w_y y_t + w_z z_t$
$= 2xs - 2ys\sin t + zs\cos t$
$= 2s^2 t - 2s^2\cos t\sin t + 2s^2\sin t\cos t$
$= 2s^2 t$

12. Let $u(x,y) = e^y F(xe^{-y^2})$.

(a) If $F(z) = \ln z$, find u_x and u_y.

(b) Show that $2xyu_x + u_y = u$.



Solution. (a) $u(x,y) = e^y F(xe^{-y^2}) = e^y \ln(xe^{-y^2}) = e^y\left(\ln x - y^2\right)$. So

$$u_x = \tfrac{1}{x}e^y \text{ and } u_y = e^y\left(\ln x - y^2 - 2y\right).$$

(b) Let $w = xe^{-y^2}$. Then

$$\begin{cases} u_x = \frac{\partial}{\partial x}(e^y F) = e^y \frac{\partial F}{\partial x} = e^y F' \frac{\partial w}{\partial x} = e^y e^{-y^2} F' = e^{y-y^2} F' & (1) \\ u_y = \frac{\partial}{\partial y}(e^y F) = e^y F + \frac{\partial F}{\partial y} = e^y F + e^y F' \frac{\partial w}{\partial y} = e^y F - e^y(2xye^{-y^2} F') = e^y F - 2xye^{y-y^2} F' & (2) \end{cases}$$

where $F = F(xe^{-y^2})$. Then

$$(1)\times 2xy + (2):\ 2xyu_x + u_y = e^y F = u.$$

13. Suppose that $w = f(xz, yz)$, where f is a differentiable function. Show that

$$xw_x + yw_y = zw_z.$$

Solution. Let $w = f(xz, yz) = f(s,t) \Rightarrow s_x = z, s_z = x, t_y = z, t_z = y$. Then

$$\begin{array}{c} f \\ /\ \backslash \\ s \quad t \\ /\ \backslash\ /\ \backslash \\ x\ z\ y\ z \end{array} \qquad \begin{aligned} w_x &= f_s s_x = f_s z \\ w_y &= f_t t_y = f_t z \\ w_z &= f_s s_z + f_t t_z = f_s x + f_t y \\ \Rightarrow xw_x + yw_y &= xzf_s + yzf_t = z\left(xf_s + yf_t\right) = zw_z \end{aligned}$$

14. Let $f(u,v)$ be a differentiable function of two variables and let z be a differentiable function of x and y defined implicitly by $f(xz(x,y), yz(x,y)) = 0$. Show that $xz_x + yz_y = -z$. Assume that $xf_u + yf_v \neq 0$.

Solution. Partially differentiate $f(u,v) = f(xz, yz) = 0$ with respect to x and y.

Note that $u_x = z, u_z = x, v_y = z, v_z = y$. If $xf_u + yf_v \neq 0$, then

$$\begin{array}{c} f \\ /\ \backslash \\ u \quad v \\ /\ \backslash\ /\ \backslash \\ x\ z\ y\ z \\ /\ \backslash\ /\ \backslash \\ x\ y\ x\ y \end{array}$$

$$0 = f_u(u_x + u_z z_x) + f_v v_z z_x = (z + xz_x)f_u + yz_x f_v \Rightarrow z_x = -\frac{zf_u}{xf_u + yf_v}$$

$$0 = f_u u_z z_y + f_v(v_y + v_z z_y) = xz_y f_u + \left(z + yz_y\right) f_v \Rightarrow z_y = -\frac{zf_v}{xf_u + yf_v}$$

$$\therefore xz_x + yz_y = -\frac{xzf_u}{xf_u + yf_v} - \frac{yzf_v}{xf_u + yf_v} = -z\frac{xf_u + yf_v}{xf_u + yf_v} = -z.$$

If $z = 0$, then obviously $xz_x + yz_y = -z$ holds true.

Now if $xf_u + yf_v = 0$ and $z \neq 0$, then

$$\begin{aligned} 0 &= zf_u + \left(xf_u + yf_v\right) z_x \\ 0 &= zf_v + \left(xf_u + yf_v\right) z_y \end{aligned} \Rightarrow zf_u = 0 = zf_v \Rightarrow f_u = 0 = f_v.$$

This means that $f(u,v) = 0$, because

$$f(u,v) = f\left(xz(x,y), yz(x,y)\right) = 0.$$

Remark: This result is obtained under the assumption of $xf_u + yf_v \neq 0$. That is equivalent, by the chain rule, to the assumption that $\frac{\partial}{\partial z} f(xz, yz) \neq 0$. That, in turn, is almost, but not quite, equivalent to the statement that $f(xz, yz) = 0$ can be solved for z as a function of x and y.

15. Let S be the surface $xy - 2x + yz + x^2 + y^2 + z^3 = 7$.

(a) The equation defining S implicitly defines z as a function of x and y for $(x,\ y,\ z)$ near $(0,\ 2,\ 1)$. Find expressions for z_x and z_y. Evaluate z_y at $(x,\ y,\ z) = (0,\ 2,\ 1)$.

(b) Find an expression for z_{xy}.

Solution. (a) Partially differentiate $xy - 2x + yz + x^2 + y^2 + z^3 = 7$ with respect to x and y :

$$y - 2 + yz_x + 2x + 3z^2 z_x = 0 \Rightarrow z_x = \tfrac{2-2x-y}{y+3z^2} \Rightarrow z_x(0,2,1) = 0$$

$$x + z + yz_y + 2y + 3z^2 z_y = 0 \Rightarrow z_y = -\tfrac{x+2y+z}{y+3z^2} \Rightarrow z_y(0,2,1) = -1$$

(b) Either find $\frac{\partial}{\partial y}\left(\frac{2-2x-y}{y+3z^2}\right)$ or $-\frac{\partial}{\partial x}\left(\frac{x+2y+z}{y+3z^2}\right)$ to obtain

$$\frac{\partial^2 z}{\partial y \partial x} = z_{xy} = -\frac{1}{y+3z^2} - \frac{2-2x-y}{(y+3z^2)^2}\left(1 - \frac{6z(x+2y+z)}{y+3z^2}\right).$$

16. Find the directional derivative of $f(x,y,z) = e^{xyz}$ in the $\underset{\sim}{j} + \underset{\sim}{k}$ direction at $(0,\ 1,\ 1)$.

Solution. Since $\nabla f = \left(yze^{xyz}, xze^{xyz}, xye^{xyz}\right)$, it follows that

$$\partial_{\underset{\sim}{j}+\underset{\sim}{k}} f = \tfrac{\underset{\sim}{j}+\underset{\sim}{k}}{\sqrt{2}} \cdot \left(yze^{xyz}, xze^{xyz}, xye^{xyz}\right) = \tfrac{1}{\sqrt{2}} x(y+z)e^{xyz} \Rightarrow \partial_{\underset{\sim}{j}+\underset{\sim}{k}} f(0,1,1) = 0.$$

Alternative. $\nabla f(0,1,1) = \left(1,0,0\right) \Rightarrow \partial_{\underset{\sim}{j}+\underset{\sim}{k}} f(0,1,1) = \frac{1}{\sqrt{2}}(0,1,1)\cdot(1,0,0) = 0$.

17. Find $\nabla\left(y^2 + \sin(xy)\right)$.

Solution. $\left(y\cos(xy), 2y + x\cos(xy), 0\right)$ if in 3-D, or without the '0' in 2-D.

18. Find $\nabla f(a,b)$ if given that $\partial_{\frac{1}{\sqrt{2}}(\underset{\sim}{i}+\underset{\sim}{j})} f(a,b) = 3\sqrt{2}$ and $\partial_{\frac{1}{5}(3\underset{\sim}{i}-4\underset{\sim}{j})} f(a,b) = 5$.

Solution. Suppose $f_x(a,b), f_y(a,b)$ exist. Let $\alpha = f_x(a,b), \beta = f_y(a,b)$. Then

$$\partial_{\frac{1}{\sqrt{2}}(\underset{\sim}{i}+\underset{\sim}{j})} f(a,b) = 3\sqrt{2} \Rightarrow \tfrac{1}{\sqrt{2}}\left(\underset{\sim}{i} + \underset{\sim}{j}\right)\cdot\left(\alpha,\beta\right) = 3\sqrt{2} \Rightarrow \alpha + \beta = 6$$

$$\partial_{\frac{1}{5}(3\underset{\sim}{i}-4\underset{\sim}{j})} f(a,b) = 5 \Rightarrow \tfrac{1}{5}\left(3\underset{\sim}{i} - 4\underset{\sim}{j}\right)\cdot\left(\alpha,\beta\right) = 5 \Rightarrow 3\alpha - 4\beta = 25$$

Solve this to get $\alpha = 7, \beta = -1$. So, $\nabla f(a,b) = \left(7,-1\right)$.

19. Let $f(x,y) = 2x^2 + 3xy + y^2$ be a function of x and y.

(a) Find the maximum rate of change of $f(x,y)$ at the point $P\left(1,-\frac{4}{3}\right)$.

(b) Find the directions vector such that the directional derivative of $f(x,y)$ at P is $\frac{1}{5}$.

Solution. (a) ∇f is the rate of change. The maximum rate of change at point $P(x,\ y)$ is given by $\left\|\nabla f(a,b)\right\|$ (norm of ∇f), i.e., $\left\|\left(4x+3y, 3x+2y\right)\right\|_{\left(1,-\frac{4}{3}\right)} = \left\|\left(0,\frac{1}{3}\right)\right\| = \frac{1}{3}$.

(b) Let $v = \left(c, \sqrt{1-c^2}\right)$ be a unit vector, then

$$\frac{1}{5} = \partial_v f(1,-\tfrac{4}{3}) = \left(c,\sqrt{1-c^2}\right)\cdot\left(0,\tfrac{1}{3}\right) = \tfrac{1}{3}\sqrt{1-c^2} \Rightarrow c = \pm\tfrac{4}{5} \Rightarrow v = \left(\pm\tfrac{4}{5},\tfrac{3}{5}\right).$$

20. Suppose that f is a function of three variables and let

$$u = \tfrac{1}{\sqrt{6}}\left(1,1,2\right), v = \tfrac{1}{\sqrt{3}}\left(1,-1,-1\right), w = \tfrac{1}{\sqrt{3}}\left(1,1,1\right).$$

Suppose that at a point $\left(a,b,c\right)$, $\partial_u f = 0 = \partial_v f = \partial_w f - 4$. Find $\nabla f(a,b,c)$.

Solution. Let $\nabla f(a,b,c) = \left(\alpha,\beta,\gamma\right)$. Then

$$\begin{aligned}
\partial_u f = 0 &\Rightarrow \tfrac{1}{\sqrt{6}}\left(1,1,2\right)\cdot\left(u,v,w\right) = 0 \Rightarrow \alpha+\beta+2\gamma = 0\\
\partial_v f = 0 &\Rightarrow \tfrac{1}{\sqrt{3}}\left(1,-1,-1\right)\cdot\left(u,v,w\right) = 0 \Rightarrow \alpha-\beta-\gamma = 0\\
\partial_w f = 4 &\Rightarrow \tfrac{1}{\sqrt{3}}\left(1,1,1\right)\cdot\left(u,v,w\right) = 4 \Rightarrow \alpha+\beta+\gamma = 4\sqrt{3}
\end{aligned}$$

This implies that $(\alpha,\beta,\gamma) = 2\sqrt{3}\left(1,3,-2\right) = \nabla f(a,b,c)$.

21. Write out the general form of the total derivative Df for the functions:

a) $f:\mathbb{R}\to\mathbb{R}$ b) $f:\mathbb{R}^3\to\mathbb{R}$ c) $f:\mathbb{R}\to\mathbb{R}^2$ d) $f:\mathbb{R}^3\to\mathbb{R}^2$

Solution.

Question	Df	Size of matrix Df
a) $f(x) = y$	$\begin{pmatrix}\frac{df}{dx}\end{pmatrix}$	1×1
b) $f(x,y,z) = w$	$\begin{pmatrix}\frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z}\end{pmatrix}$	1×3
c) $f(x) = (f_1(x), f_2(x))$	$\begin{pmatrix}\frac{df_1}{dx}\\ \frac{df_2}{dx}\end{pmatrix}$	2×1
d) $f(x,y,z) = (f_1(x,y,z), f_2(x,y,z))$	$\begin{pmatrix}\frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z}\\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z}\end{pmatrix}$	2×3

22. Write out the total derivative Dg for the following functions:

a) $g(x,y,z) = xyz$ b) $g(x,y) = \begin{pmatrix} x^2+y^3 \\ \cos(xy) \end{pmatrix}$ c) $g(x) = \begin{pmatrix} x^2 \\ x^3 \end{pmatrix}$ d) $g(x,y,z) = \begin{pmatrix} xyz \\ x^2+y^3+z^4 \end{pmatrix}$

Function	Dg	Size of Dg
a) $g:\mathbb{R}^3 \to \mathbb{R}$ $g(x,y,z) = xyz$	$\begin{pmatrix} yz & xz & xy \end{pmatrix}$	1×3
b) $g:\mathbb{R}^2 \to \mathbb{R}^2$ $g(x,y) = \begin{pmatrix} x^2+y^3 \\ \cos(xy) \end{pmatrix}$	$\begin{pmatrix} 2x & 3y^2 \\ -y\sin(xy) & -x\sin(xy) \end{pmatrix}$	2×2
c) $g:\mathbb{R} \to \mathbb{R}^2$ $g(x) = \begin{pmatrix} x^2 \\ x^3 \end{pmatrix}$	$\begin{pmatrix} 2x \\ 3x^2 \end{pmatrix}$	2×1
d) $g:\mathbb{R}^3 \to \mathbb{R}^2$ $g(x,y,z) = \begin{pmatrix} xyz \\ x^2+y^3+z^4 \end{pmatrix}$	$\begin{pmatrix} yz & xz & xy \\ 2x & 3y^2 & 4z^3 \end{pmatrix}$	2×3

23. Find the total derivative matrix Dh of the function h defined as follows for part (a) and (b) of the previous question.

(a) $h(x,y,z) = f(g(x,y,z))$ where $f(u) = \begin{pmatrix} u^2 \\ u \end{pmatrix}$.

(b) $h(x,y) = f(g(x,y))$ where $f(u,v) = \begin{pmatrix} uv \\ v \end{pmatrix}$.

Solution. (a) $h(x,y,z) = f(g(x,y,z)) = f(xyz) = \begin{pmatrix} (xyz)^2 \\ xyz \end{pmatrix} : \mathbb{R}^3 \to \mathbb{R}^2$. So,

$$Dh = \begin{pmatrix} 2x(yz)^2 & 2y(xz)^2 & 2z(xy)^2 \\ yz & xz & xy \end{pmatrix}.$$

(b) $h(x,y) = f(g(x,y)) = f\begin{pmatrix} x^2+y^3 \\ \cos(xy) \end{pmatrix} = \begin{pmatrix} (x^2+y^3)\cos(xy) \\ \cos(xy) \end{pmatrix} : \mathbb{R}^2 \to \mathbb{R}^2$. So,

$$Dh = \begin{pmatrix} 2x\cos(xy) - y(x^2+y^3)\sin(xy) & 3y^2\cos(xy) - x(x^2+y^3)\sin(xy) \\ -y\sin(xy) & -x\sin(xy) \end{pmatrix}.$$

24. Define $f:\mathbb{R}^n \to \mathbb{R}^n$ by $f(x)=x$ (identity function on $\mathbb{R}^n$), $x \in \mathbb{R}^n$. Show that, for any $a \in \mathbb{R}^n$, $Df(a)=I_n=$ identity matrix of size n by n.

Proof. Let $x=(x_1,...,x_n) \in \mathbb{R}^n$ where $x_i \in \mathbb{R}\ \forall i=1,...,n$. Write $f(x)=x$ as

$$f(x_1,...,x_n)=(x_1,...,x_n)=(f_1(x),...,f_n(x))=(f_1(x_1,...,x_n),...,f_n(x_1,...,x_n)).$$

Then we see that $f_i(x_1,...,x_n)=x_i \forall i=1,...,n$. Therefore,

$$\frac{\partial f_i}{\partial x_j}=\begin{cases}1, i=j\\ 0, i\neq j\end{cases},\ i,\ j=1,\ \ldots,\ n.$$

Hence, the total derivative (which is a square matrix)

$$Df=\begin{pmatrix}\frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n}\\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n}\\ \vdots & \vdots & \ddots & \vdots\\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n}\end{pmatrix} \in M_n(\mathbb{R}) \Rightarrow Df=\begin{pmatrix}1 & 0 & \cdots & 0\\ 0 & 1 & \ddots & \vdots\\ \vdots & \ddots & \ddots & 0\\ 0 & \cdots & 0 & 1\end{pmatrix}=I_n.$$

We see that for any point $a=(a_1,...,a_n)$, $Df(a)=I_n$.

Past years test/final exam questions

25. Let $z(x,y):\mathbb{R}^2\to\mathbb{R}$ be a function satisfying $x^2+(y-z)^2=1$ and $z(x,y)\neq y$, where x and y are independent variables.

(a) Show that $z_{xy}(x,y)=z_{yx}(x,y)$ (without using the mixed derivative theorem).

(b) Hence, evaluate $z_{xy}(\pi,-1)$.

Solution. $x^2+(y-z)^2=1$

(a) $\underset{\frac{\partial}{\partial y}}{\Rightarrow} 2(y-z)(1-z_y)=0\Rightarrow z_y=1\because z(x,y)\neq y\Rightarrow z_{yx}=0^{\#}$

$\underset{\frac{\partial}{\partial x}}{\Rightarrow} 2x-2(y-z)z_x=0\ ^*\Rightarrow z_x=\frac{x}{y-z}, z\neq y$

From*, we partially differentiate with respect to y:

$(1-z_y)z_x+(y-z)z_{xy}=0\Rightarrow z_{xy}=\frac{(1-z_y)z_x}{z-y}=0\because z_y=1, z\neq y$

$\therefore z_{xy}=0=z_{yx}$ from $^{\#}$

(b) $z_{xy}(\pi,-1)=0$ in fact $z_{xy}(x,y)=0$ for any x and y.

26. Let $f(u,v)$ be a differentiable function of two variables, and let z be a differentiable function of x and y defined implicitly by $f\left(\frac{z}{x},\frac{z}{y}\right)=0$, $x,y\neq 0$. Given that $x^{-1}f_u+y^{-1}f_v\neq 0$, prove that $xz_x+yz_y=z$.

Proof. Let $u=x^{-1}z, v=y^{-1}z$. Given that $x,y\neq 0$ and $x^{-1}f_u+y^{-1}f_v\neq 0$.

$$\begin{cases} u_x=-x^{-2}z+x^{-1}z_x \\ u_y=x^{-1}z_y \end{cases} \qquad \begin{cases} v_x=y^{-1}z_x \\ v_y=-y^{-2}z+y^{-1}z_y \end{cases} \ \dagger$$

† or by symmetry, $x\leftrightarrow y$, $u\leftrightarrow v$.

$0=f_u\cdot u_x+f_v\cdot v_x=f_u\cdot\left(-x^{-2}z+x^{-1}z_x\right)+f_v\cdot y^{-1}z_x$

$\Rightarrow x^{-1}zf_u=x\left(x^{-1}f_u+y^{-1}f_v\right)z_x\Rightarrow xz_x=\frac{x^{-1}zf_u}{x^{-1}f_u+y^{-1}f_v}$

$0=f_u\cdot u_y+f_v\cdot v_y=f_u\cdot x^{-1}z_y+f_v\cdot\left(-y^{-2}z+y^{-1}z_y\right)$ ‡

$\Rightarrow y^{-1}zf_v=y\left(x^{-1}f_u+y^{-1}f_v\right)z_y\Rightarrow yz_y=\frac{y^{-1}zf_v}{x^{-1}f_u+y^{-1}f_v}$

Hence, $xz_x+yz_y=\frac{x^{-1}f_u+y^{-1}f_v}{x^{-1}f_u+y^{-1}f_v}z=z$.

‡ or by symmetry, immediately arrive at the conclusion.

27. Let $f:\mathbb{R}^3\to\mathbb{R}$ be a differentiable function. Let $u=\frac{3}{5}\underset{\sim}{i}+\frac{4}{5}\underset{\sim}{j}$, $v=\frac{3}{5}\underset{\sim}{i}-\frac{4}{5}\underset{\sim}{j}$ and $w=\frac{3}{5}\underset{\sim}{i}+\frac{4}{5}\underset{\sim}{k}$ be vectors and given that the directional derivatives at point (a,b,c) along the directions of vectors u, v and w are given as follows:

$$\begin{cases}\partial_u f(a,b,c)=10\\ \partial_v f(a,b,c)=2\\ \partial_w f(a,b,c)=14.\end{cases}$$

Find the maximum rate of change of f at point (a,b,c).

Solution. Let $\nabla f\left(a,b,c\right)=\left(x,y,z\right)$.

$$\begin{cases}D_u f\left(a,b,c\right)=10\\ D_v f\left(a,b,c\right)=2\\ D_w f\left(a,b,c\right)=14\end{cases}\Rightarrow\begin{cases}\frac{3}{5}x+\frac{4}{5}y=10\\ \frac{3}{5}x-\frac{4}{5}y=2\\ \frac{3}{5}x+\frac{4}{5}z=14\end{cases}\text{ implies that } x=10,\ y=5,\ z=10$$

$$\therefore\left|\nabla f\left(a,b,c\right)\right|=5\sqrt{4+1+4}=15$$

28. Let $f:\mathbb{R}^2\to\mathbb{R}$ be a function of u and v, where $u,v:\mathbb{R}^2\to\mathbb{R}$ are the functions of (x,y) given by $\left(u,v\right)=\left(xy,\frac{x}{y}\right)$. Let $h(x,y)=f(u(x,y),v(x,y))$.

(a) Find h_x and h_y.

(b) Show that $xh_x+yh_y=2uf_u$ and $xh_x-yh_y=2vf_v$.

Solution. (a) First note that $y\neq 0$. Then

$$h_x=f_u u_x+f_v v_x=f_u\times y+f_v\times\frac{1}{y}=yf_u+y^{-1}f_v$$
$$h_y=f_u u_y+f_v v_y=f_u\times x+f_v\times\left(-\frac{x}{y^2}\right)=xf_u-xy^{-2}f_v$$

(b) Hence, $xh_x+yh_y=xyf_u+xy^{-1}f_v+xyf_u-xy^{-1}f_v=2xyf_u=2uf_u$, and

$xh_x-yh_y=xyf_u+xy^{-1}f_v-xyf_u+xy^{-1}f_v=2xy^{-1}f_v=2vf_v$.

29. (a) Let $f:\mathbb{R}\to\mathbb{R}$ be twice differentiable and $h(x,y)=f(x)f(y^2)+f(xy)$.

(i) Find the expressions for h_{xy} in terms of f' and f''.

(ii) Hence, evaluate $h_{xy}(1,1)$ if given that $f'(1)=f''(1)=1$.

(b) Let $g:\mathbb{R}^2\to\mathbb{R}$ be a function defined as $g(u,v)=u^2+v^2$ with $u:\mathbb{R}^2\to\mathbb{R}$ and $v:\mathbb{R}\times\mathbb{R}^+\to\mathbb{R}$ defined as $u(x,y)=xe^y$, $v(x,y)=x\ln y$. Write $\frac{\partial g}{\partial x}$ and $\frac{\partial g}{\partial y}$ in terms of x and y.

Solution. (a) (i) $h_x=f'(x)f(y^2)+yf'(xy)\Rightarrow h_{xy}=2yf'(x)f'(y^2)+f'(xy)+xyf''(xy)$

(ii) $h_{xy}(1,1)=2+1+1=4$

(b) $\frac{\partial g}{\partial x}=g_u u_x+g_v v_x=2ue^y+2v\ln y=2xe^{2y}+2x(\ln y)^2$

$\frac{\partial g}{\partial y}=g_u u_y+g_v v_y=2uxe^y+2v\frac{x}{y}=2x^2e^{2y}+2\left(\frac{x^2}{y}\right)\ln y$

30. Let $f,x,y:\mathbb{R}^2\to\mathbb{R}$ be functions differentiable in $\mathbb{R}^2$. Let $(u,v)\in\mathbb{R}^2$. Define

$$g(u,v)=f(x(u,v),y(u,v)).$$

(a) State the chain rule for $\frac{\partial g}{\partial u}$.

(b) Suppose that $x(u,v)=2u+3v$ and $y(u,v)=v-u^3$. Find the formula for $\frac{\partial g}{\partial u}$ and the formula for $\frac{\partial^2 g}{\partial u\partial v}$ in terms of $f_x,f_y,f_{xx},f_{xy},f_{yy},u$ and v.

Solution. (a) $\frac{\partial g}{\partial u}=\frac{\partial f}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial f}{\partial y}\frac{\partial y}{\partial u}$.

(b) $\begin{cases}x_u=2\\x_v=3\end{cases}$ and $\begin{cases}y_u=-3u^2\\y_v=1\end{cases}$ implies that $\frac{\partial g}{\partial u}=2f_x-3u^2f_y$, and

$$\begin{aligned}\frac{\partial^2 g}{\partial u\partial v}&=\frac{\partial}{\partial v}\left(\frac{\partial g}{\partial u}\right)\\&=\frac{\partial}{\partial v}\left(2f_x-3u^2f_y\right)\\&=2\frac{\partial}{\partial v}f_x-3u^2\frac{\partial}{\partial v}f_y\\&=2(f_{xx}x_v+f_{xy}y_v)-3u^2(f_{yx}x_v+f_{yy}y_v)\\&=2(3f_{xx}+f_{xy})-3u^2(3f_{xy}+f_{yy})\\&=6f_{xx}+(2-9u^2)f_{xy}-3u^2f_{yy}\end{aligned}$$